

Question	Marking Guidance	Mark	Comments
06.1	1. Extract DNA and add restriction endonucleases/restriction enzymes; 2. Separate fragments using electrophoresis; 3. (Treat DNA to) form single strands OR (Treat DNA to) expose bases; 4. The probe will bind to/hybridise/base pair with the <i>SUT1</i> /gene; 5. Use autoradiography (to show the bound probe);	4 max	3. Ignore method used to separate strands 5. Accept use photographic or X ray film (to show the bound probe) 5. X rays alone is not sufficient

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06.2	1. Antisense mRNA is <u>complementary</u> to 'sense' mRNA; 2. Antisense mRNA would bind/base pair to (sense) mRNA; OR Double stranded (m)RNA forms; 3. Ribosomes would not be able to bind; 4. Preventing/less translation (of mRNA) OR Preventing/less production of SUT1 (protein);	4	4. Accept descriptions of translation

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06.3	0.4 (318):1;	1	Accept any suitable rounding

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06.4	1. Some ($^{14}\text{CO}_2$) used to make cellulose/cell walls; 2. Some ($^{14}\text{CO}_2$) converted into starch (which remains in the leaf); 3. Not all ($^{14}\text{CO}_2$) fixed/used in photosynthesis; OR Not enough RuBP (to combine with all of the $^{14}\text{CO}_2$); 4. Some used to reform RuBP OR Some (is still) in glycerate 3-phosphate/GP/triose phosphate/in the Calvin cycle;	2 max	1. Accept some becomes lipids/ proteins/DNA/RNA/ nucleotides 2. Accept some ($^{14}\text{CO}_2$) converted into glucose 3. Accept descriptions of this

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06.5	1. Reduced <i>SUT1</i> expression/less SUT 1 (protein) means less sucrose exported (so concentration increases in leaves); 2. (Increased sucrose in leaves) inhibits rubisco, so less $^{14}\text{CO}_2$ fixed into GP; OR (Increased sucrose in leaves) inhibits rubisco, so less $^{14}\text{CO}_2$ combines with RuBP; OR (Increased sucrose in leaves) inhibits rubisco, so less Calvin cycle/light <u>in</u>dependent reaction/s; 3. Less sucrose transported to roots, so roots do not develop/grow (as shown by larger shoot to root dry mass ratio); 4. Roots less developed so fewer minerals available for growth 5. Less growth means less dry mass;	4 max	2. Accept less rubisco or less active rubisco for 'inhibits rubisco' 4. Accept: roots less developed so less water available for photosynthesis 5. Accept: less photosynthesis/light <u>in</u>dependent reaction/s means less dry mass;